SECTION A – 1 MARK QUESTIONS (10 × 1 = 10 Marks)

1. What is the main function of a vector database?

a) Store large text files
b) Store and efficiently search high-dimensional vectors
c) Encrypt data
d) Manage database transactions

2. How does RAG reduce hallucinations in LLMs?

a) Uses larger models
b) Grounds responses in retrieved verified documents
c) Increases training data
d) Uses more parameters

3. Which API is commonly used for building chatbots with generative AI?

a) REST API for databases
b) OpenAI API, Anthropic API, Google Gemini API
c) Socket.io
d) GraphQL

4. What is semantic similarity in AI applications?

a) Similarity in spelling
b) Similarity in meaning captured by embeddings
c) Similarity in word count
d) Similarity in language

5. Which tool is used for semantic search implementation?

a) SQL databases
b) File systems
c) Vector databases with embeddings
d) Hash tables

6. What is multimodal AI?

a) Multiple training runs
b) Processing multiple data types (text, image, audio) together
c) Multiple models in sequence
d) Multiple programming languages

7. How is context injected into prompts in RAG?

a) Direct text concatenation
b) Retrieved documents added to augment the prompt
c) Database queries
d) Network requests

8. What is the purpose of the retrieval component in RAG?

a) Train the model
b) Find relevant knowledge from external database
c) Compress data
d) Encrypt communications

9. Which of these is a use case for multimodal AI?

a) Image captioning
b) Visual question answering
c) Medical diagnosis from images and text
d) All of above

10. What is a chatbot fine-tuned on domain data likely to improve?

a) Inference speed
b) Model size
c) Domain-specific accuracy and relevance
d) Training cost

SECTION B – 5 MARK QUESTIONS Answer ANY 6 (6 × 5 = 30 Marks)

1. Explain attention mechanism in transformers.

2. What is few-shot learning?

3. Define vector search.

4. What is model evaluation?

5. Explain safe AI deployment.

6. Write short notes on Generative Adversarial Networks (GANs).

7. What is semantic similarity in AI?

SECTION C – 10 MARK QUESTIONS Answer ANY 4 (4 × 10 = 40 Marks)

1. Describe LoRA and PEFT techniques.

2. Explain AI guardrails and filters.

3. Discuss logging and monitoring in AI systems.

4. Explain multimodal AI systems with examples.

5. Discuss AI bias and fairness in model training.

SECTION C – 20 MARK CASE STUDY Answer ANY 1 (1 × 20 = 20 Marks)

CASE STUDY 1: EDUCATIONAL AI TUTOR SYSTEM

SCENARIO:

EduMind develops an AI tutor system for K-12 students to provide personalized learning. The system uses multimodal AI to teach various subjects. Challenges include:

AI provides incorrect explanations in complex topics

Diverse learning needs not addressed equally across student demographics

Privacy concerns regarding student learning data collection

Dependency on AI reducing critical thinking and problem-solving skills

Parental concerns about screen time and digital safety

QUESTIONS

Q1 Design an evaluation framework ensuring accuracy through expert validation and identifying topics where AI struggles most.

Q2 Propose adaptive learning mechanisms addressing diverse student needs and ensuring equitable educational outcomes.

Q3 Create a privacy-preserving architecture complying with COPPA (Children's Online Privacy Protection Act) and educational data laws.

Q4 Design human-in-the-loop mechanisms where teachers guide AI and students develop critical thinking alongside AI assistance.

CASE STUDY 2: CUSTOMER SERVICE CHATBOT DEPLOYMENT

SCENARIO:

RetailGiant deploys an LLM-powered customer service chatbot handling support across 50 countries. After launch:

Chatbot misunderstands customer issues, providing irrelevant solutions

AI leaks sensitive customer information (order details, payment info) in responses

Multilingual support quality varies significantly across languages

AI generates tone-deaf responses offending cultural sensibilities

No escalation path: Frustrated customers unable to reach human agents

QUESTIONS

Q1 Design RAG-based retrieval system grounding responses in company policies and FAQs while preventing data leakage.

Q2 Implement content moderation and PII (Personally Identifiable Information) redaction ensuring customer data protection.

Q3 Develop multilingual quality assurance framework evaluating cultural appropriateness and response accuracy per language.

Q4 Create intelligent escalation system routing complex issues to human agents with context preservation and SLA tracking.